MAT3032 ANSWERS TO EXERCISES 2.4

Exercises 2.4

1. Under the conjugation action of G on itself, $\mathcal{O}(x) = \{gxg^{-1} \; : \; g \in G\}$ which is the conjugacy class of x. $\text{Fix}(g) = \{x \in G \; : \; gxg^{-1} = x\} = C_G(g)$.

   By Burnside's formula, number of conjugacy classes = number of orbits = $\dfrac{1}{|G|}\sum_{g\in G} |\text{Fix}(g)| = \dfrac{1}{|G|}\sum_{g\in G} |C_G(g)|$.

   For S_3, the conjugacy classes of $\iota, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau$ have orders 6, 3, 3, 2, 2, 2 respectively, and $\dfrac{1}{6}(6+3+3+2+2+2) = 3$.

2. The symmetries are the identity, two reflections (i.e. turning the hexagon over) and a 180° rotation. The symmetries form a 4-element subgroup of S_6, isomorphic to D_2 or V.

   (a) Labelling $A, \ldots, F$ as $1, \ldots, 6$, the symmetries, as permutations in S_6, are $(1)(2)(3)(4)(5)(6)$, $(1\;\;5)(2\;\;4)(3)(6)$, $(1\;\;2)(4\;\;5)(3\;\;6)$, $(1\;\;4)(2\;\;5)(3\;\;6)$, having cycle indices 6, 4, 3, 3.

   Hence the number of q-colourings is $\dfrac{1}{4}(q^6 + q^4 + 2q^3)$.

   (b) Labelling the edges $1, \ldots, 6$, the symmetries have the same cycle types as in (a), giving the same result.

   (c) With suitable labelling of the triangles as $1, \ldots, 6$ the symmetries are $(1)(2)(3)(4)(5)(6)$, $(1\;\;2)(5\;\;6)(3)(4)$, $(1\;\;2)(3\;\;4)(5\;\;6)$, $(3\;\;4)(1)(2)(5)(6)$, having cycle indices 6, 4, 3, 5 so the number of q-colourings is
   $\dfrac{1}{4}(q^6 + q^5 + q^4 + q^3)$.

3. (a) There are five symmetries in the plane: the identity and four 5-cycles, with cycle types 5, 1, 1, 1, 1, so the number of q-colourings is $\dfrac{1}{5}(q^5 + 4q)$.

   (b) There are ten symmetries if rotations out of the plane (equivalent to reflection) are allowed: the identity, four 5-cycles and five relections of the same type as $(1)(2\;\;5)(3\;\;4)$, with cycle types 5, 1, 1, 1, 1, 3, 3, 3, 3, 3.

   so the number of q-colourings is $\dfrac{1}{10}(q^5 + 5q^3 + 4q)$.

4. $|A_4| = 12$. Its elements are the even permutations in S_4: eight 3-cycles of the form $(a\;\;b\;\;c)(d)$, with cycle index 2; three products of two disjoint transpositions $(a\;\;b)(c\;\;d)$, with cycle index 2, and the identity $(1)(2)(3)(4)$, with cycle index 4. Thus the number of 4-colourings is $\dfrac{1}{12}(4^4 + 11(4^2)) = 36$.

   Suppose each colour is used once. The identity fixes $4! = 24$ colourings and the other symmetries fix none, so the number is $\dfrac{1}{12}(24+0) = 2$.

5. There are six symmetries of the triangular prism.

   (a) Label the triangular faces 1, 2 and the rectangular faces 3, 4, 5.

   The symmetries, as permutations in S_5, are $(1)(2)(3)(4)(5)$, $(1)(2)(3\ \ 4\ \ 5)$, $(1)(2)(3\ \ 5\ \ 4)$, $(1\ \ 2)(3\ \ 4)(5)$, $(1\ \ 2)(3\ \ 5)(4)$, $(1\ \ 2)(3)(4\ \ 5)$. These have cycle indices 5, 3, 3, 3, 3, 3 so the number of q-colourings is $\frac{1}{6}(q^5 + 5q^3)$.

   (b) Label the vertices 1, 2, 3 around one triangular face and 4, 5, 6 round the other, so that 1 is joined to 4, etc.

   The symmetries, as permutations in S_6, are $(1)(2)(3)(4)(5)(6)$, $(1\ 2\ 3)(4\ 5\ 6)$, $(1\ \ 3\ \ 2)(4\ \ 6\ \ 5)$, $(1\ \ 4)(2\ \ 6)(3\ \ 5)$, $(1\ \ 6)(2\ \ 4)(3\ \ 5)$, $(1\ \ 5)(2\ \ 4)(3\ \ 6)$. These have cycle indices 6, 2, 2, 3, 3, 3 so the number of q-colourings is $\frac{1}{6}(q^6 + 3q^3 + 2q^2)$.

   (c) Label the edges 1, 2, 3 around one triangular face, 4, 5, 6 round the other, and 7, 8, 9 along the lengths of the prism.

   The symmetries, as permutations in S_9, are $(1)(2)(3)(4)(5)(6)(7)(8)(9)$, $(1\ \ 2\ \ 3)(4\ \ 5\ \ 6)(7\ \ 8\ \ 9)$, $(1\ \ 3\ \ 2)(4\ \ 6\ \ 5)(7\ \ 9\ \ 8)$, $(1\ \ 5)(2\ \ 4)(3\ \ 6)(7)(8\ \ 9)$, $(1\ \ 4)(2\ \ 6)(3\ \ 5)(8)(7\ \ 9)$, $(1\ \ 6)(2\ \ 5)(3\ \ 4)(9)(7\ \ 8)$. These have cycle indices 9, 3, 3, 5, 5, 5 so the number of q-colourings is $\frac{1}{6}(q^9 + 3q^5 + 2q^3)$.

6. Label the edges 1 to 4 around one square end, 5 to 8 around the other square end, and 9 to 12 for the long edges. Then the eight symmetries are:

   $(1)(2)(3)(4)(5)(6)(7)(8)(9)(10)(11)(12)$, $(1\ \ 2\ \ 3\ \ 4)(5\ \ 6\ \ 7\ \ 8)(9\ \ 10\ \ 11\ \ 12)$,

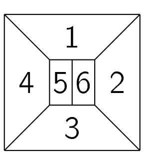
   $(1\ \ 3)(2\ \ 4)(5\ \ 7)(6\ \ 8)(9\ \ 11)(10\ \ 12)$, $(1\ \ 4\ \ 3\ \ 2)(5\ \ 8\ \ 7\ \ 6)(9\ \ 12\ \ 11\ \ 10)$,

   $(1\ \ 5)(2\ \ 8)(3\ \ 7)(4\ \ 6)(9\ \ 10)(11\ \ 12)$, $(1\ \ 6)(2\ \ 5)(3\ \ 8)(4\ \ 7)(9\ \ 11)(10)(12)$,

   $(1\ \ 7)(2\ \ 6)(3\ \ 5)(4\ \ 8)(9\ \ 12)(10\ \ 11)$, $(1\ \ 8)(2\ \ 7)(3\ \ 6)(4\ \ 5)(10\ \ 12)(9)(11)$,

   so the number of q-colourings is $\frac{1}{8}(q^{12} + 2q^7 + 3q^6 + 2q^3)$.

7. The q^6 term shows that there are six regions. The fraction $\frac{1}{4}$ shows that there are four symmetries. The figure could be:

   in which the symmetries are $(1)(2)(3)(4)(5)(6)$, $(2\ \ 4)(5\ \ 6)(1)(3)$, $(1\ \ 3)(2)(4)(5)(6)$, $(1\ \ 3)(2\ \ 4)(5\ \ 6)$. These are all self-inverse, so the symmetry group is a subgroup of S_6 that is isomorphic to V (or D_2).